

Preface

Who is This Book For?

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Why Write Yet Another Book About Linear Algebra?

The direct answer to this question is simple: I really think most resources are doing a *bad job* of explaining fundamental linear algebra to new students ¹. Not because they lack material or the complete picture - but because the common approach to teaching fundamental mathematical topics just doesn't work well for most people ².

Introduction courses to fundamental mathematical topics usually look like this: here're some basic definitions, let's derive some specific results from them - now go apply these in your field.

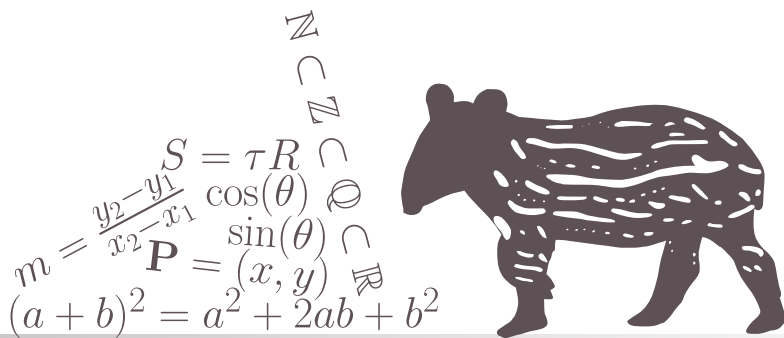
¹and honestly, this isn't restricted to linear algebra but applies to all fundamental mathematical topics - but it's a classic case

²the main exception is students of pure mathematics and people who can easily deal with extremely abstract concepts - a very specific group of people at the end of the day

CHAPTER

1

THE BASICS OF LINEAR ALGEBRA

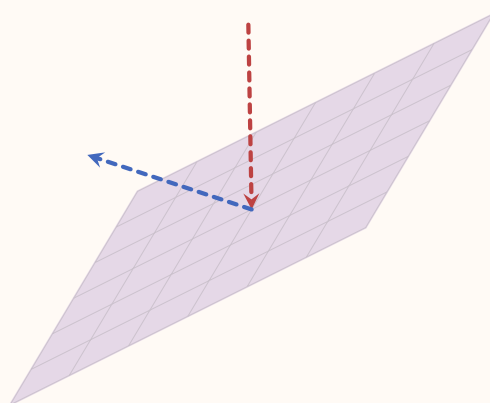


1.1 VECTORS AND VECTOR SPACES

1.1.1 Motivation

Motivating Problem 1.1 Bouncing/Reflecting Off a Wall

Imagine we are writing a 3D computer game in which some object is bouncing off a wall. Alternatively, consider 3D graphics in general, where one simulates rays of light reflecting off of smooth surfaces. In both cases the same problem needs to be solved: given a flat plane of some orientation in space, and an object hitting it from some direction - what would be the direction of the object after it bounces off the plane?



To solve this problem we will need to know how to represent quantities like position, velocity and orientation in 3D, and more generally points, lines and planes - and to know how to calculate the intersection between these different objects. By the end of this section you should be able to solve this problem by yourself, and even generalize it to any number of dimensions.



Motivating Problem 1.2 Gravity Force

Consider a simple simulation of classical (“Newtonian”) gravity, say of a planet with mass m orbiting a star with mass M (where $M \gg m$). At any given moment, the force acting on the planet due to the star’s gravity has magnitude

$$F = G \frac{Mm}{r^2},$$

where r is the distance between the planet and the star at that moment, and G is a universal constant (specifically, $G = 6.674 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$).

However, since we live in 3-dimensional space^a, we also want this force to be pointing in the direction from the planet to the star. To do this, we need to understand how to describe orientations in 3-dimensional space, and be able to pack together both orientation and magnitude in a correct way.

^aallegedly.



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CHAPTER 1. THE BASICS OF LINEAR ALGEBRA

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1.1. VECTORS AND VECTOR SPACES

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1.1.2 Geomteric Properties of Vectors

1.1.3 Common Coordinate Systems in 2- and 3-Dimensions

1.1.4 Geomteric Objects

1.1.5 Projections and Rejections

1.1.6 Vector Spaces

1.1.7 Solving the Motivating Problems

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